FYS5419/9419, lecture May 7

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Brief reminder from last week

Basics of support vector machines (without a margin)

$$\mathcal{L} = \sum_{i=1}^{n} \lambda_i - \frac{1}{2} \sum_{ij} \lambda_i \lambda_j y_i y_j K(\vec{x}_i, \vec{x}_j)$$

linear kernel $K(\vec{x}_i, \vec{x}_j) = K_{ij}$

$$= \vec{x}_i^T \vec{x}_j \qquad h_i = y_i \lambda_i$$

$$\sum_{i=1}^{n} h_i h_j K_{ij} \geq 0 \quad , \text{ then}$$

for each set of n ($\alpha_i \in \mathbb{R}$)
then there is a mapping
$\phi(\vec{x})$ such that — K —
can be written as

$$K_{ij} = \phi^T(\vec{x_i})\,\phi(\vec{x_j})$$

$$\left(\sum_{i=1}^{n} \alpha_i \phi^T(\vec{x_i}) \right) \left(\sum_{j=1}^{n} \alpha_j \phi(\vec{x_j}) \right)$$

Feature function/mapping

$$f(\vec{x}) = \sum_{i=1}^{n} \alpha_i K(\phi(\vec{x_i}), \phi(\vec{x}))$$

quantum mechanical equivalent

$$x_i \rightarrow |\phi(\vec{x}_i)\rangle$$

$$\rho(x) = |\phi(x)\rangle\langle\phi(x)|$$

$$k(\vec{x}_i, \vec{x}_j) = k_{ij} = Tr\{\rho(\vec{x}_i)\rho(\vec{x}_j)\}$$

$$= |\langle\phi(\vec{x}_i)|\phi(x_j)\rangle|^2$$

$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$R_x(x) = e^{-i\frac{x}{2}\sigma_x}$$

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

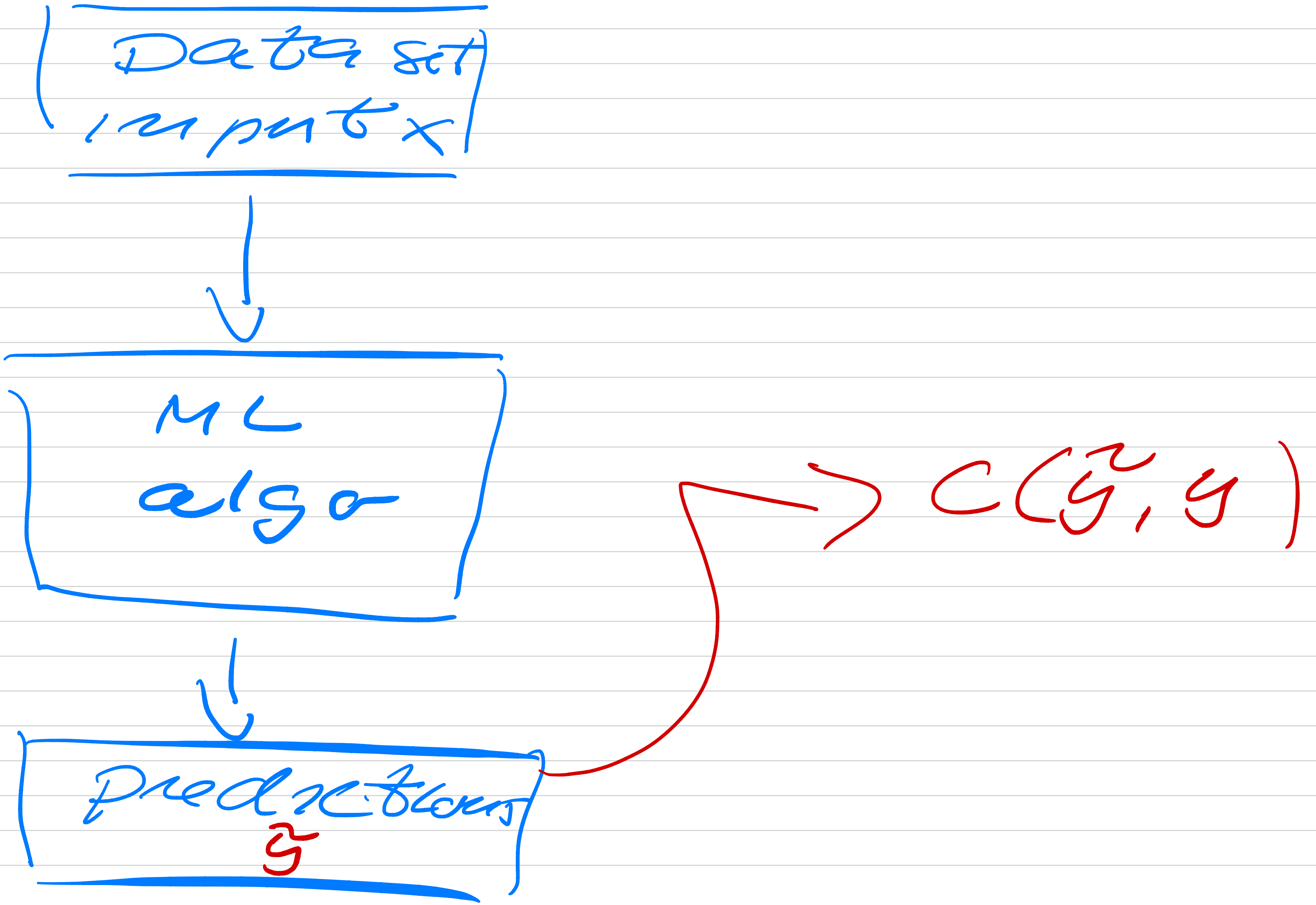
$$R_x(x) = \cos\left(\frac{x}{2}\right)\mathbb{1} - i\sin\left(\frac{x}{2}\right)\sigma_x$$

$$R_x(x)|0\rangle = |\phi(x)\rangle$$

$$\rho(x) = |\phi(x)\rangle\langle\phi(x)|$$

$$k(x, x') = \left(\cos\left(\frac{x - x'}{2}\right)\right)^2$$

Data encoding (classical)



Quantum model

Data set
x, y

↓

Encoding — quantum state preparation

↓

QML algorithm — parameterized circuit (unitary transform)

↓

measurement → Readout → Predictions

Parameterized QC

user

problem/model and input/training data

classical comp
initial data
pre-processing

post processing
parameter update
(gradients

setup PQC

Quantum comp
running quantum circuit
state preparation and measurement

measurement

result

Four possible classes / measurement

0	$\|00\rangle$	100
1	$\|01\rangle$	550
2	$\|10\rangle$	200
3	$\|11\rangle$	150

Two qubits

$$|00\rangle = |0\rangle \otimes |0\rangle$$
$$|01\rangle = |0\rangle \otimes |1\rangle$$

Encoding of data

$$|\psi\rangle = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\phi}\sin\left(\frac{\theta}{2}\right) \end{bmatrix}$$

$\theta \in [0, \pi]$ $\phi \in [0, 2\pi]$

8-feature dataset with N samples

all $x_1, \ldots x_8$ are real-valued

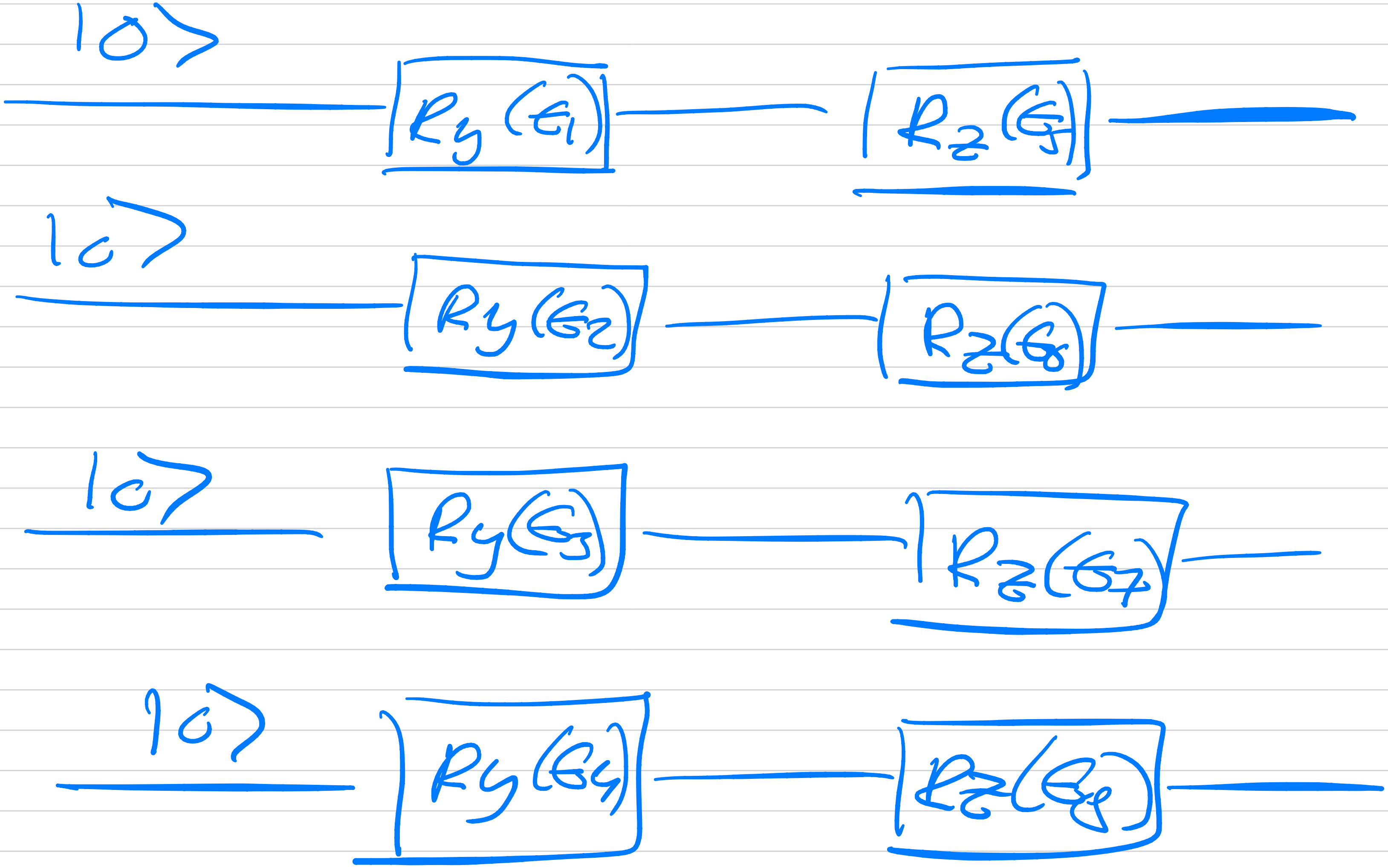
$i = 1, 2, \ldots 8$

x_i^{min} and x_i^{max}

For every sample $j = 1, 2, \ldots N$
we can establish a one-to-
one correspondence

$$\theta_i^j = \frac{x_i^j - x_i^{min}}{x_i^{max} - x_i^{min}} \pi$$

$$x_i^{min} = \min_j x_i^j \qquad x_i^{max} = \max_j x_i^j$$


|0⟩
Ry(θ1)
Rz(θ5)
|0⟩
Ry(θ2)
Rz(θ6)
|0⟩
Ry(θ3)
Rz(θ7)
|0⟩
Ry(θ4)
Rz(θ8)

$$x^j \mapsto \bigotimes_{i=1}^{K} \left(\cos\left(\frac{\theta_i^j}{2}\right)|0\rangle + \sin\left(\frac{\theta_i^j}{2}\right)|1\rangle \right)$$

$$x^j = (x_1^j \; x_2^j \; \ldots \; x_8^j) \in \mathbb{R} \qquad j = 1, 2, \ldots N$$

Basis encoding

assume we have a binary dataset D where each pattern $x^m \in D$ is a binary string

$$x^m = (b_1^m, \ldots \; b_N^m] \quad b_i^m \in \{0, 1\}$$

$x^1 = (00, 01) \rightarrow x^1 = 0001$

$x^2 = (11, 10) \rightarrow x^2 = 1110$

and represent in terms of basis states

$|0001\rangle$ and $|1110\rangle$

The superposition is

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0001\rangle + |1110\rangle\right)$$

$2^4 = 16$

$$|0001\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|x\rangle_4 = |x_0 x_1 x_2 x_3\rangle$$

$$x = x_0 + 2x_1 + 4x_2 + 8x_3$$

$x_0 = 0 \quad x_1 = 0, \quad x_2 = 0 \quad x_3 = \underline{1}$

$x = 8$, state

$|x\rangle_4 = |1000\rangle$ state $\underline{1}$

$|x\rangle_4 = |1110\rangle \rightarrow x = 7$

state 7

$$|\psi\rangle = \frac{1}{\sqrt{\underset{\shortparallel\,D}{2^4}}} \sum_{i=1}^{D} \alpha_i |x_i\rangle_4$$

$$\alpha^T = \left[0, 0, 0, 0, 0, 0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, - - 0\right] \qquad (2^4 - 1)$$